5

MONTE CARLO METHODS FOR LEARNING

In this chapter, we show how sampling methods can be utilised for statistical learning of parameters in latent variable models.

Another application where sampling methods come in handy is *model learning* problems. So far, we have covered probabilistic inference problems where we typically introduced a prior $p(x)$ and a likelihood $p(y|x)$. Our sampling algorithms usually handled *unnormalised* likelihoods – therefore, we could just set $\bar{p}_\star(x) = p(y|x)p(x)$ and run the sampling algorithm (say MCMC). However, in many cases, we are interested in learning the parameters of the model. In fact, in fields like generative models (e.g. energy-based models), the density of interest could be completely parameterised with a flexible neural network (as we will see in the next section). In this section, we will see such methods and general principles to learn parametric probabilistic models from data and the use of stochastic simulation methods for this task.

5.1 MAXIMUM MARGINAL LIKELIHOOD ESTIMATION

Let us start, from the simple Bayesian inference scenario. Assume that, we have a prior $p_{\theta_1}(x)$ and a likelihood $p_{\theta_2}(y|x)$, where θ_1 and θ_2 are the parameters of the prior and the likelihood, respectively. Let $\theta = (\theta_1, \theta_2)$ and we will typically from now on write the joint density $p_\theta(x, y)$ to leave the dependence on θ explicit. Assume that y is observed and fixed data. Then, the maximum marginal likelihood estimation (MMLE) problem is defined as

$$\theta^\star = \arg\max_\theta \log p_\theta(y) = \arg\max_\theta \log \int p_\theta(x, y)\mathrm{d}x.$$

We note that, despite its conceptual similarity, this is a very different problem than a pure maximum likelihood estimation (MLE) problem. In MLE, we would have $\log p_\theta(y)$ available and we could treat this as a simple optimisation problem. In contrast, the MMLE problem above involves an intractable integral. This is suitable for the earlier methods we introduced in this course, as in this case, we need to solve the integration and optimisation problem together.

Example 5.1. Consider the following model

$$p_\theta(x) = \mathcal{N}(x; \theta, \sigma_0^2),$$
$$p(y|x) = \mathcal{N}(y; ax, \sigma^2).$$

where $a \neq 0$ is a known constant and $\sigma_0^2, \sigma^2 > 0$ are known variances. Given the observation y, derive the MMLE solution for θ.

Solution. We can write the marginal likelihood as

$$p_\theta(y) = \int p(y|x) p_\theta(x) \mathrm{d}x,$$
$$= \mathcal{N}(y; a\theta, a^2\sigma_0^2 + \sigma^2).$$

Computing the $\log p_\theta(y)$, we have

$$\log p_\theta(y) = -\frac{1}{2}\log(2\pi(a^2\sigma_0^2 + \sigma^2)) - \frac{(y - a\theta)^2}{2(a^2\sigma_0^2 + \sigma^2)}.$$

Compute now

$$\frac{\partial \log p_\theta(y)}{\partial \theta} = \frac{a(y - a\theta)}{a^2\sigma_0^2 + \sigma^2}.$$

By setting this to zero, we obtain $\theta^\star = y/a$.

5.1.1 THE EXPECTATION-MAXIMISATION ALGORITHM

A standard method to learn parameters in the presence of hidden, latent, or unobserved variables is the Expectation-Maximisation (EM) algorithm. The EM algorithm is a well-established algorithm, first proposed by Dempster et al. (1977) and widely studied in several forms. In this text, we will introduce the main template of the algorithm and prove the ascend property to show that the EM algorithm can maximise the likelihood (in the absence of local minima).

The EM algorithm aims at producing iterates $(\theta_t)_{t\geq 0}$ that converge to a local maximum of the marginal likelihood. Given a parameter estimate θ_{t-1}, the EM algorithm is a two-step iterative algorithm that alternates between two steps to obtain the next estimate:

1. **Expectation step (E-step):** Compute the expected value of the log likelihood with respect to the posterior distribution of the latent variables: $Q(\theta, \theta_{t-1}) = \mathbb{E}_{p_{\theta_{t-1}}(x|y)}[\log p_\theta(x, y)]$.

2. **Maximisation step (M-step):** Maximise the expected log likelihood with respect to the parameters: $\theta_t = \arg\max_\theta Q(\theta, \theta_{t-1})$.

More compactly, we can write the two steps in a single line:

$$\theta_t \in \operatorname*{argmax}_\theta \mathbb{E}_{p_{\theta_{t-1}}(x|y)}[\log p_\theta(x,y)]. \tag{5.1}$$

The main challenges of implementing this method is obvious: E-step is usually intractable, as we have seen throughout this course. $Q(\theta, \theta_{t-1})$ is an integral with respect the posterior $p_{\theta_{t-1}}(x|y)$ which cannot be usually found in closed form (and even usually not possible to sample i.i.d). Similarly, the maximisation step for $Q(\theta, \theta_{t-1})$ (or its approximations) is intractable. We can easily prove, however, the base scheme *does* maximise the likelihood.

Proposition 5.1 (EM algorithm ascends the likelihood). *The EM algorithm is guaranteed to ascend the likelihood, i.e., the likelihood at each iteration is greater or equal to the previous iteration, i.e.,* $\log p_{\theta_t}(y) \geq \log p_{\theta_{t-1}}(y)$.

Proof. We write:

$$\begin{aligned}
\log p_\theta(y) &= \log \int p_\theta(x,y)\mathrm{d}x \\
&= \log \int \frac{p_{\theta_k}(x,y)}{p_{\theta_k}(x,y)} p_\theta(x,y)\mathrm{d}x \\
&= \log p_{\theta_k}(y) + \log \int p_{\theta_k}(x|y) \frac{p_\theta(x,y)}{p_{\theta_k}(x,y)}\mathrm{d}x \\
&\geq \log p_{\theta_k}(y) + \underbrace{\int p_{\theta_k}(x|y) \log p_\theta(x,y)\mathrm{d}x - \int p_{\theta_k}(x|y)\log p_{\theta_k}(x,y)\mathrm{d}x}_{\Delta(\theta,\theta_k)}
\end{aligned}$$

where

$$\Delta(\theta,\theta_k) = Q(\theta,\theta_k) - Q(\theta_k,\theta_k). \tag{5.2}$$

In other words, let $\theta_{k+1} \in \arg\max_\theta Q(\theta,\theta_k)$. Then

$$\log p_{\theta_{k+1}}(y) \geq \log p_{\theta_k}(y) + \underbrace{Q(\theta_{k+1},\theta_k) - Q(\theta_k,\theta_k)}_{\geq 0}$$

Ascent is achieved by this method. □

!! **Example 5.2.** Consider again

$$\begin{aligned}
p_\theta(x) &= \mathcal{N}(x;\theta,\sigma_0^2), \\
p(y|x) &= \mathcal{N}(y;ax,\sigma^2),
\end{aligned}$$

with $\sigma_0^2, \sigma^2 > 0$ known and y observed. Assume $a \neq 0$. Derive the EM recursions for this model and show that EM converges to the MMLE solution.

Solution. At iteration k, the E-step computes $p_{\theta_k}(x|y)$. By Gaussian conjugacy,

$$p_{\theta_k}(x|y) = \mathcal{N}(x; m_k, s^2),$$

where

$$s^2 = \left(\frac{1}{\sigma_0^2} + \frac{a^2}{\sigma^2}\right)^{-1} = \frac{\sigma_0^2\sigma^2}{\sigma^2 + a^2\sigma_0^2},$$
$$m_k = s^2\left(\frac{\theta_k}{\sigma_0^2} + \frac{ay}{\sigma^2}\right) = \frac{\sigma^2\theta_k + a\sigma_0^2 y}{\sigma^2 + a^2\sigma_0^2}.$$

The joint log-likelihood is

$$\log p_\theta(x, y) = C(y, x) - \frac{(x-\theta)^2}{2\sigma_0^2},$$

so

$$\begin{aligned} Q(\theta, \theta_k) &= \mathbb{E}_{p_{\theta_k}(x|y)}[\log p_\theta(x, y)] \\ &= C_k - \frac{1}{2\sigma_0^2}\mathbb{E}_{p_{\theta_k}(x|y)}[(x-\theta)^2] \\ &= C_k - \frac{1}{2\sigma_0^2}\left((\theta - m_k)^2 + s^2\right), \end{aligned}$$

for a constant C_k independent of θ. Hence, the M-step yields

$$\theta_{k+1} = \arg\max_\theta Q(\theta, \theta_k) = m_k = \frac{\sigma^2}{\sigma^2 + a^2\sigma_0^2}\theta_k + \frac{a\sigma_0^2}{\sigma^2 + a^2\sigma_0^2}y.$$

Define

$$r := \frac{\sigma^2}{\sigma^2 + a^2\sigma_0^2} \in (0, 1), \qquad b := \frac{a\sigma_0^2}{\sigma^2 + a^2\sigma_0^2}y.$$

Then EM is the affine recursion

$$\theta_{k+1} = r\theta_k + b.$$

We can then write, starting from θ_0,

$$\theta_{k+1} = r^{k+1}\theta_0 + b\sum_{i=0}^{k} r^i = r^{k+1}\theta_0 + b\frac{1 - r^{k+1}}{1 - r},$$

since $r \neq 1$. Taking the limit as $k \to \infty$ and using $|r| < 1$, we have $r^{k+1} \to 0$ and $1 - r^{k+1} \to 1$. Hence, We obtain then

$$\lim_{k\to\infty} \theta_k = \frac{b}{1-r} = \frac{a\sigma_0^2}{\sigma^2 + a^2\sigma_0^2}y\frac{\sigma^2 + a^2\sigma_0^2}{a^2\sigma_0^2} = \frac{y}{a} = \theta^\star.$$

5.1.2 GRADIENT BASED OPTIMISATION FOR MMLE

As mentioned above, E and M steps of the EM algorithm are usually intractable. Therefore, a common approach is to combine sampling methods to approximate the E-step and gradient-based optimisation methods to approximate the M-step. We turn now to the gradient-based optimisation methods for MMLE, and will clarify the connection between the method below and the EM algorithm at the end of this section.

Assume that we would like to maximise $\log p_\theta(y)$ w.r.t. θ using a gradient-based method. Ideally, this method would look like:

$$\theta_{t+1} = \theta_t + \delta \nabla_\theta \log p_{\theta_t}(y) \tag{5.3}$$

where $\delta > 0$ is the step size. The difficulty is that, we cannot perform standard optimisation here, as $\log p_\theta(y)$ in general not available as $p_\theta(y)$ is given as an integral. Therefore, a direct computation of $\nabla \log p_\theta(y)$ is not possible. Luckily, we can prove the following result.

Proposition 5.2 (Fisher's identity). *The gradient of the log marginal likelihood can be written as*

$$\nabla_\theta \log p_\theta(y) = \int p_\theta(x|y) \nabla_\theta \log p_\theta(x, y) \mathrm{d}x,$$

in other words, this gradient can be written as an expectation:

$$\nabla_\theta \log p_\theta(y) = \mathbb{E}_{p_\theta(x|y)}[\nabla_\theta \log p_\theta(x, y)].$$

Proof. The proof of this result is simple. We first use the fact that

$$\nabla_\theta \log p_\theta(y) = \frac{\nabla_\theta p_\theta(y)}{p_\theta(y)}.$$

Then, we can substitute $p_\theta(y) = \int p_\theta(x, y) dx$ and exchange differentiation and integration

$$\begin{aligned}
\nabla_\theta \log p_\theta(y) &= \frac{\nabla_\theta \int p_\theta(x, y) dx}{\int p_\theta(x, y) dx}, \\
&= \frac{\int \nabla_\theta p_\theta(x, y) dx}{\int p_\theta(x, y) dx}, \\
&= \frac{\int \nabla_\theta \log p_\theta(x, y) p_\theta(x, y) dx}{\int p_\theta(x, y) dx}, \\
&= \int \nabla_\theta \log p_\theta(x, y) p_\theta(x|y) dx,
\end{aligned}$$

where we used the fact that $p_\theta(x|y) = \frac{p_\theta(x,y)}{p_\theta(y)}$ in the last line. □

This formula is particularly useful as it provides a recipe to approximate the gradient of the log marginal likelihood. Assume that $(x_k)_{k=1}^K$ are (approximate) samples from $p_\theta(x|y)$, then we can approximate the gradient as

$$\nabla_\theta \log p_\theta(y) = \mathbb{E}_{p_\theta(x|y)}[\nabla_\theta \log p_\theta(x, y)] \approx \frac{1}{K} \sum_{k=1}^K \nabla_\theta \log p_\theta(x_k, y).$$

Algorithm 14 Stochastic Optimisation via Unadjusted Langevin (SOUL)

Input: Initial parameter θ_0, step size $\delta > 0$, Langevin step size $\gamma > 0$, number of Langevin steps K, burn-in steps B, $X_0^{(M)}$.
for $t = 1, \ldots, T$ **do**
  Initialise $X_t^{(0)} = X_{t-1}^{(M)}$.
  for $m = 1, \ldots, M$ **do**
    $X_t^{(m)} = X_t^{(m-1)} + \gamma \nabla_x \log p_{\theta_{t-1}}(X_t^{(m-1)}, y) + \sqrt{2\gamma} Z_k$, where $Z_k \sim \mathcal{N}(0, I)$.
  end for
  $\theta_t = \theta_{t-1} + \delta \frac{1}{M-B} \sum_{m=B+1}^{M} \nabla_\theta \log p_{\theta_{t-1}}(X_t^{(m)}, y)$.
end for

We summarise below one full algorithm to realise this method, called stochastic optimisation via unadjusted Langevin (SOUL) method (De Bortoli et al., 2021).

5.1.3 Stochastic optimisation via unadjusted Langevin (SOUL)

As summarised above, we would like to implement the gradient ascent procedure given in Eq. (5.3) where the gradient is approximated with samples from $p_\theta(x|y)$. The SOUL method is a simple method to implement this procedure by running an unadjusted Langevin dynamics to sample from $p_\theta(x|y)$ and then using these samples to approximate the gradient.

Assume that we have at the iteration θ_{t-1} and we would like to obtain θ_t. Using Fisher's identity, we can see that our gradient at θ_{t-1} can be written as an expectation

$$\nabla_\theta \log p_{\theta_{t-1}}(y) = \mathbb{E}_{p_{\theta_{t-1}}(x|y)}[\nabla_\theta \log p_{\theta_{t-1}}(x, y)].$$

Therefore, at iteration θ_{t-1}, we need to sample from $p_{\theta_{t-1}}(x|y)$ to approximate this gradient. We can run an unadjusted Langevin dynamics targeting $p_{\theta_{t-1}}(x|y)$ to obtain these samples. In particular, we can run the following dynamics for $m = 1, \ldots, M$:

$$X_t^{(m)} = X_t^{(m-1)} + \gamma \nabla_x \log p_{\theta_{t-1}}(X_t^{(m-1)}, y) + \sqrt{2\gamma} Z_k, \tag{5.4}$$

where $\gamma > 0$ is the step size and $Z_k \sim \mathcal{N}(0, I)$. $X_t^{(0)}$ can be initialised with any value, but a common choice is to use the last sample from the previous iteration, i.e., $X_t^{(0)} = X_{t-1}^{(M)}$. After running this dynamics for M steps, we can use the samples after the burn-in (B) iterations to approximate the gradient as

$$\nabla_\theta \log p_{\theta_{t-1}}(y) \approx \frac{1}{M-B} \sum_{m=B+1}^{M} \nabla_\theta \log p_{\theta_{t-1}}(X_t^{(m)}, y).$$

This can then be used for the gradient ascent step to obtain θ_t as

$$\theta_t = \theta_{t-1} + \delta \frac{1}{M-B} \sum_{m=B+1}^{M} \nabla_\theta \log p_{\theta_{t-1}}(X_t^{(m)}, y).$$

The full algorithm is summarised in Algorithm 14.

The issue with this algorithm is that it is computationally expensive as we need to run a long Langevin dynamics at each iteration. Motivated by this, several particle-based versions have been developed which take the perspective of joint sampling of (x, θ) with a single interacting particle SDE system, see, e.g., Kuntz et al. (2023); Akyildiz et al. (2025) for recent developments in this direction.

Remark 5.1. Obviously, this is a generic recipe to perform gradient-based optimisation for MMLE problems. We can replace the unadjusted Langevin dynamics in (5.4) with any other sampling method we have seen so far, e.g., MH, MALA, Gibbs, SGLD, or more advanced methods like Hamiltonian Monte Carlo (HMC) etc. Similarly, we can replace the gradient ascent step with any other optimisation method, e.g., Adam, RMSProp, etc. The main point is that we need to use the samples from $p_{\theta_{t-1}}(x|y)$ to approximate the gradient and then use this approximation for the optimisation step.

5.2 BAYESIAN ESTIMATION OF PARAMETERS

In this section, we will also put priors on parameters θ. This is a very common approach in Bayesian statistics and machine learning, as it allows us to incorporate prior knowledge about the parameters and also to quantify uncertainty about the parameters. To be specific, we will consider the following model. We have a prior on parameters, $p(\theta)$, and on latent variables (potentially dependent on θ), $p(x|\theta)$. We also have a likelihood $p(y|x, \theta)$[1].

In this setting, we are primarily interested in sampling from $p(\theta|y)$ for parameter estimation. However, this is not possible in general as $p(\theta|y)$ is given as an integral

$$p(\theta|y) = \int p(\theta, x|y)\mathrm{d}x,$$

where

$$p(\theta, x|y) = \frac{p(y|x, \theta)p(x|\theta)p(\theta)}{p(y)}.$$

In the following sections, we describe a couple of ways to sample from $p(\theta|y)$.

5.2.1 SAMPLING FROM THE POSTERIOR $p(\theta, x|y)$

A simple way to sample from $p(\theta|y)$ is to sample from the joint posterior $p(\theta, x|y)$ and then marginalise out x. This can be done with any of the sampling methods we have seen so far. For example, we can run a Metropolis-Hastings algorithm targeting $p(\theta, x|y)$ and then use the θ-samples to approximate the posterior $p(\theta|y)$. Similarly, we can run a Langevin dynamics targeting $p(\theta, x|y)$ and then use the samples of θ to approximate the posterior $p(\theta|y)$. We can also run a Gibbs sampler by alternating between sampling θ given x and sampling x given θ. The main point is that we need to sample from the joint posterior $p(\theta, x|y)$ and then use these samples to approximate the posterior $p(\theta|y)$.

As an example, we provide the Metropolis-Hastings algorithm for sampling from $p(\theta, x|y)$ in Algorithm 15.

[1] We moved θ from being in the subscript to the argument of the likelihood to make the dependence on θ explicit. In other words, we write $p(y|x, \theta)$ instead of $p_\theta(y|x)$ and $p(y|\theta)$ instead of $p_\theta(y)$ and so on.

Algorithm 15 Metropolis-Hastings for sampling from $p(\theta, x|y)$

1: **Input:** Initial parameter θ_0, initial latent variable x_0, number of iterations T.
2: **for** $t = 1, \dots, T$ **do**
3: Propose $(\theta', x') \sim q(\theta', x'|\theta_{t-1}, x_{t-1})$.
4: Compute the acceptance probability

$$\alpha = \min\left(1, \frac{p(y|x', \theta')p(x'|\theta')p(\theta')q(\theta_{t-1}, x_{t-1}|\theta', x')}{p(y|x_{t-1}, \theta_{t-1})p(x_{t-1}|\theta_{t-1})p(\theta_{t-1})q(\theta', x'|\theta_{t-1}, x_{t-1})}\right).$$

5: With probability α, set $(\theta_t, x_t) = (\theta', x')$, otherwise set $(\theta_t, x_t) = (\theta_{t-1}, x_{t-1})$.
6: **end for**

Remark 5.2. The joint sampling idea can be computationally expensive when x is high-dimensional as we need to sample from the joint posterior $p(\theta, x|y)$ which can be very complex. Moreover, the acceptance probability can be very low if the proposal distribution is not well-designed, which can lead to slow convergence. Therefore, it is important to design good proposal distributions that can efficiently explore the joint posterior $p(\theta, x|y)$.

5.2.2 PSEUDO-MARGINAL METROPOLIS-HASTINGS

Another way to sample from $p(\theta|y)$ is to use pseudo-marginal MCMC methods. The idea is to construct an unbiased estimator of the marginal likelihood $p(y|\theta)$ and then use this estimator in a Metropolis-Hastings algorithm to sample from $p(\theta|y)$. This method is particularly useful when the marginal likelihood is intractable but we can obtain an unbiased estimator of it, e.g., with importance sampling. The main advantage of this method is that it allows us to sample from $p(\theta|y)$ without having to sample from the joint posterior $p(\theta, x|y)$, which can be computationally tricky when x is high-dimensional. However, the main challenge of this method is that the variance of the estimator can be very high, which can lead to slow convergence. Therefore, it is important to design good estimators of the marginal likelihood that have low variance.

The basis of this method is the fact that the MH algorithm remains valid if one replaces $\bar{p}_\star(\cdot)$ in the usual acceptance ratio with its unbiased estimate $\widehat{p}_\star(\cdot)$ (Andrieu and Roberts, 2009). This version of the method is termed pseudo-marginal Metropolis-Hastings (PMMH). This is particularly useful when $\bar{p}_\star(\cdot)$ is intractable to evaluate, which will be our setting.

Consider the problem that we would like to sample from $p(\theta|y) \propto p(y|\theta)p(\theta)$. In our setting, the marginal likelihood $p(y|\theta)$ is given as an integral

$$p(y|\theta) = \int p(y|x, \theta)p(x|\theta)dx,$$

For any θ, we know that we can estimate this quantity in an unbiased way using importance sampling (IS) since we can evaluate $p(y|x,\theta)p(x|\theta)$ pointwise, e.g.

$$\begin{aligned} p(y|\theta) &= \int p(y|x,\theta)p(x|\theta)dx, \\ &= \int \frac{p(y|x,\theta)p(x|\theta)}{q(x|\theta)} q(x|\theta)dx, \\ &= \mathbb{E}_{q(x|\theta)} \left[\frac{p(y|x,\theta)p(x|\theta)}{q(x|\theta)} \right], \end{aligned}$$

where $q(x|\theta)$ is any proposal distribution. Therefore, we can construct an unbiased estimator of $p(y|\theta)$ as

$$\widehat{p(y|\theta)} = \frac{1}{N} \sum_{i=1}^{N} \frac{p(y|X_i,\theta)p(X_i|\theta)}{q(X_i|\theta)},$$

where $X_i \sim q(x|\theta)$ i.i.d. (we kept θ dependence but we didn't have to). It is easy to see that $\mathbb{E}[\widehat{p(y|\theta)}] = p(y|\theta)$. The acceptance rate in the algorithm in this case would take the form

$$\widehat{\alpha} = \min\left(1, \frac{\widehat{p(y|\theta')}p(\theta')q(\theta_{t-1}|\theta')}{\widehat{p(y|\theta_{t-1})}p(\theta_{t-1})q(\theta'|\theta_{t-1})}\right).$$

The full algorithm is summarised in Algorithm 16 below. The following result guarantees the correctness of this method.

Proposition 5.3 (Correctness of PMMH)**.** *Assume that for each θ we can construct a non-negative estimator $\widehat{p(y|\theta)} = g(y,\theta,u)$ using auxiliary randomness $U \sim m_\theta(\cdot)$ such that*

$$\mathbb{E}_{U\sim m_\theta}\left[g(y,\theta,U)\right] = p(y|\theta).$$

Define the extended target on (θ, u) by

$$\widetilde{p_\star}(\theta,u) \propto p(\theta)\, m_\theta(u)\, g(y,\theta,u).$$

Then the MH kernel with proposal $q(\theta'|\theta)m_{\theta'}(u')$ and acceptance probability

$$\widehat{\alpha}\Big((\theta,u),(\theta',u')\Big) = \min\left(1, \frac{g(y,\theta',u')p(\theta')q(\theta|\theta')}{g(y,\theta,u)p(\theta)q(\theta'|\theta)}\right)$$

leaves $\widetilde{p_\star}$ invariant. Moreover, the θ-marginal of $\widetilde{p_\star}$ is $p(\theta|y)$. Hence PMMH is exact for sampling from $p(\theta|y)$.

Proof. For the augmented target $\widetilde{p_\star}(\theta,u)$ and proposal

$$r\Big((\theta',u') \mid (\theta,u)\Big) = q(\theta'|\theta)m_{\theta'}(u'),$$

Algorithm 16 Pseudo-Marginal Metropolis-Hastings (PMMH)

1: **Input:** Initial pair (θ_0, u_0) with $u_0 \sim m_{\theta_0}(\cdot)$, or equivalently an initial estimate $\widehat{p(y \mid \theta_0)} = g(y, \theta_0, u_0)$, and number of iterations T.
2: **for** $t = 1, \ldots, T$ **do**
3: Propose $\theta' \sim q(\theta' | \theta_{t-1})$.
4: Draw $u' \sim m_{\theta'}(\cdot)$ and compute $\widehat{p(y|\theta')} = g(y, \theta', u')$ (e.g. with N importance samples).
5: Compute the acceptance probability

$$\widehat{\alpha} = \min\left(1, \frac{\widehat{p(y|\theta')}p(\theta')q(\theta_{t-1}|\theta')}{\widehat{p(y|\theta_{t-1})}p(\theta_{t-1})q(\theta'|\theta_{t-1})}\right).$$

6: With probability $\widehat{\alpha}$, set $(\theta_t, u_t) = (\theta', u')$, otherwise set $(\theta_t, u_t) = (\theta_{t-1}, u_{t-1})$.
7: **end for**

the standard MH acceptance probability is

$$\widehat{\alpha}\Big((\theta, u), (\theta', u')\Big) = \min\left(1, \frac{\widetilde{p_\star}(\theta', u')\, r\Big((\theta, u) \mid (\theta', u')\Big)}{\widetilde{p_\star}(\theta, u)\, r\Big((\theta', u') \mid (\theta, u)\Big)}\right),$$

with reverse proposal $r\Big((\theta, u) \mid (\theta', u')\Big) = q(\theta|\theta')m_\theta(u)$. Substituting $\widetilde{p_\star}(\theta, u) \propto p(\theta)m_\theta(u)g(y, \theta, u)$ gives

$$\begin{aligned}\widehat{\alpha}\Big((\theta, u), (\theta', u')\Big) &= \min\left(1, \frac{p(\theta')m_{\theta'}(u')g(y, \theta', u')\, q(\theta|\theta')m_\theta(u)}{p(\theta)m_\theta(u)g(y, \theta, u)\, q(\theta'|\theta)m_{\theta'}(u')}\right) \\ &= \min\left(1, \frac{g(y, \theta', u')p(\theta')q(\theta|\theta')}{g(y, \theta, u)p(\theta)q(\theta'|\theta)}\right),\end{aligned}$$

since $m_\theta(u)$ and $m_{\theta'}(u')$ cancel exactly. Therefore this kernel is precisely a standard MH kernel on the augmented space, and standard MH detailed balance gives invariance of $\widetilde{p_\star}$. Reusing the previous likelihood estimate means that the current state is the pair (θ, u), so the denominator $g(y, \theta, u)$ is held fixed during the proposal step rather than being resampled. If the proposal is rejected then the chain remains at the same pair (θ, u), whereas if it is accepted the new pair (θ', u') becomes the current state. This induces serial correlation across iterations, but this is exactly the dependence structure of the MH chain on the augmented space and does not affect invariance. For the marginal,

$$\int \widetilde{p_\star}(\theta, u)\, du \propto p(\theta) \int m_\theta(u)g(y, \theta, u)\, du = p(\theta)\, p(y|\theta) \propto p(\theta|y),$$

where unbiasedness was used in the middle equality. □

Remark 5.3. The PMMH algorithm is a powerful method for sampling from $p(\theta|y)$ when the marginal likelihood is intractable. However, the main challenge of this method is that the variance of the estimator $\widehat{p(y|\theta)}$ can be very high, which can lead to slow convergence. Therefore, it is important to design good estimators of the marginal likelihood that have low variance. One common approach to reduce the variance is to use control variates or to use more sophisticated proposal distributions for the importance sampling step.

5.3 SOLVED EXAMPLES

Example 5.3 (Exact PMMH acceptance under a two-point likelihood-noise model)**.** Assume a symmetric proposal $q(\theta'|\theta) = q(\theta|\theta')$. Suppose the likelihood estimator has multiplicative noise

$$\widehat{p(y|\theta)} = p(y|\theta)\,U, \qquad U = \begin{cases} 1+c, & \text{with probability } 1/2, \\ 1-c, & \text{with probability } 1/2, \end{cases} \qquad 0 < c < 1.$$

Then $\mathbb{E}[U] = 1$, so $\widehat{p}(y|\theta)$ is unbiased. Show how the mean PMMH acceptance probability changes with c and how it compares to the original MH acceptance probability α.

Solution. Let $g(y, \theta, u) = p(y|\theta)u$ and $m(u)$ be the distribution of U. Then the extended target is

$$\widetilde{p_\star}(\theta, u) \propto p(\theta)m(u)g(y, \theta, u) = p(\theta)m(u)p(y|\theta)u.$$

At stationarity on the extended space, the PMMH acceptance probability is

$$\begin{aligned}
\widehat{\alpha} &= \min\left(1, \frac{\widehat{p(y|\theta')}p(\theta')}{\widehat{p(y|\theta)}p(\theta)}\right) \\
&= \min\left(1, \frac{p(y|\theta')U'p(\theta')}{p(y|\theta)Up(\theta)}\right) \\
&= \min\left(1, r\,\frac{U'}{U}\right), \qquad r := \frac{p(y|\theta')p(\theta')}{p(y|\theta)p(\theta)} = \frac{p(\theta'|y)}{p(\theta|y)}.
\end{aligned}$$

Hence the original MH acceptance probability is

$$\alpha = \min(1, r).$$

Moreover at stationarity, we have

why?

$$p(u) \propto m(u)u,$$

Because p(y) is a constant with respect to both theta and u, it gets absorbed into the proportionality constant.

$$p(u) \propto m(u) u \underbrace{\int p(\theta) p(y|\theta) \, d\theta}_{p(y)}$$

means that the current noise U is distributed as

$$\mathbb{P}(U = 1+c) = \frac{1+c}{2}, \qquad \mathbb{P}(U = 1-c) = \frac{1-c}{2},$$

while the proposed noise satisfies

$$\mathbb{P}(U' = 1 + c) = \mathbb{P}(U' = 1 - c) = \frac{1}{2}.$$

Then the four possible (U', U) pairs have probabilities

$$\mathbb{P}(1+c, 1+c) = \frac{1+c}{4}, \qquad \mathbb{P}(1-c, 1+c) = \frac{1+c}{4},$$
$$\mathbb{P}(1+c, 1-c) = \frac{1-c}{4}, \qquad \mathbb{P}(1-c, 1-c) = \frac{1-c}{4}.$$

These stationary pair probabilities induce

$$\frac{U'}{U} \in \left\{1, \frac{1+c}{1-c}, \frac{1-c}{1+c}\right\},$$

with probabilities $1/2$, $\frac{1-c}{4}$, $\frac{1+c}{4}$ respectively.

Now define

$$r := \frac{p(\theta'|y)}{p(\theta|y)}, \qquad \alpha := \min(1, r), \qquad \kappa_+ := \frac{1+c}{1-c}, \qquad \kappa_- := \frac{1-c}{1+c}.$$

The stationary conditional mean PMMH acceptance is

$$\bar{\alpha}(r, c) := \mathbb{E}[\hat{\alpha} \mid \theta, \theta'] = \frac{1}{2}\min(1, r) + \frac{1+c}{4}\min(1, \kappa_- r) + \frac{1-c}{4}\min(1, \kappa_+ r).$$

Equivalently, let us consider the following cases.

1. Assume $r \geq \kappa_+$. In this case, the original MH acceptance rate $\alpha = 1$. Also $\kappa_+ r \geq \kappa_- r \geq 1$, so

   $$\bar{\alpha}(r, c) = \frac{1}{2} + \frac{1}{4} + \frac{1}{4} = 1.$$

   Therefore, PMMH does not degrade accept/reject discrimination for very good moves, i.e., for $r \geq \frac{1+c}{1-c}$.

2. Assume $1 \leq r < \kappa_+$. This is still the regime that the original acceptance rate $\alpha = 1$ i.e. the original sampler accepts at this regime. Note now that

   $$\kappa_- r < \kappa_- \kappa_+ = 1,$$

   and $\kappa_+ r \geq 1$ (since both κ_+ and r are ≥ 1), so

   $$\bar{\alpha}(r, c) = \frac{1}{2} + \frac{1+c}{4}\kappa_- r + \frac{1-c}{4} = \frac{3-c}{4} + \frac{1-c}{4} r.$$

   Moreover,

   $$\alpha - \bar{\alpha}(r, c) = 1 - \left(\frac{3-c}{4} + \frac{1-c}{4} r\right)$$
   $$= \frac{1 + c - (1-c)r}{4} > 0,$$

   because $r < \kappa_+ = \frac{1+c}{1-c}$. Hence $\bar{\alpha}(r, c) < \alpha = 1$. Thus, PMMH starts rejecting some moves that the original MH would accept.

3. Assume $\kappa_- \le r < 1$. Then $\alpha = \min(1, r) = r$. Also, $\kappa_- r < 1$, and $\kappa_+ r \ge \kappa_+ \kappa_- = 1$ (since $r \ge \kappa_-$), so

$$\begin{aligned}\bar{\alpha}(r, c) &= \frac{1}{2}r + \frac{1+c}{4}\kappa_- r + \frac{1-c}{4} \\ &= \frac{1-c}{4} + \frac{3-c}{4}r.\end{aligned}$$

Indeed,

$$\bar{\alpha}(r, c) - \alpha = \frac{1 - c - (1+c)r}{4} \le 0,$$

with equality if and only if $r = \kappa_- = \frac{1-c}{1+c}$. Hence $\bar{\alpha}(r, c) \le \alpha$ in this regime, with equality only at the lower boundary $r = \kappa_-$. Thus, in this intermediate region, PMMH has a smaller mean acceptance probability than exact MH.

4. Assume $0 \le r < \kappa_-$. Then $\alpha = r$, $\kappa_- r < 1$, and $\kappa_+ r < 1$, so

$$\bar{\alpha}(r, c) = \frac{1}{2}r + \frac{1+c}{4}\kappa_- r + \frac{1-c}{4}\kappa_+ r = r = \alpha.$$

In this regime, the mean PMMH acceptance matches the original MH acceptance.

In summary, the behaviour relative to the exact MH acceptance α is:

$$\bar{\alpha}(r, c) = \begin{cases} 1, & r \ge \kappa_+, \\ \dfrac{3-c}{4} + \dfrac{1-c}{4}r, & 1 \le r < \kappa_+, \\ \dfrac{1-c}{4} + \dfrac{3-c}{4}r, & \kappa_- \le r < 1, \\ r, & r < \kappa_-. \end{cases}$$

Thus PMMH behaves exactly like MH for very good moves ($r \ge \kappa_+$) and very bad moves ($r \le \kappa_-$). Noise only affects the intermediate region $\kappa_- < r < \kappa_+$.

Example 5.4 (EM vs exact gradient ascent in the same Gaussian model)**.** Consider Example 5.1 with $a \neq 0$. Compare EM with exact gradient ascent on $\log p_\theta(y)$.

Solution. From the previous example, EM has recursion

$$\theta_{k+1}^{\mathrm{EM}} = r_{\mathrm{EM}}^{k+1}\theta_0 + b\frac{1 - r_{\mathrm{EM}}^{k+1}}{1 - r_{\mathrm{EM}}},$$

where

$$r_{\mathrm{EM}} := \frac{\sigma^2}{\sigma^2 + a^2\sigma_0^2}, \qquad b := \frac{a\sigma_0^2}{\sigma^2 + a^2\sigma_0^2} y.$$

Recall that

$$\frac{b}{1 - r_{\mathrm{EM}}} = \frac{y}{a} = \theta^\star,$$

On the other hand, the exact gradient ascent with step size $\delta > 0$ has recursion

$$\begin{aligned}
\theta_{k+1}^{\mathrm{GD}} &= \theta_k^{\mathrm{GD}} + \delta \nabla_\theta \log p_\theta(y)\Big|_{\theta=\theta_k^{\mathrm{GD}}} \\
&= \theta_k^{\mathrm{GD}} + \delta \frac{a}{\sigma^2 + a^2\sigma_0^2}(y - a\theta_k^{\mathrm{GD}}) \\
&= r_{\mathrm{GD}}\theta_k^{\mathrm{GD}} + \delta \frac{a}{\sigma^2 + a^2\sigma_0^2} y,
\end{aligned}$$

where

$$r_{\mathrm{GD}} := 1 - \delta \frac{a^2}{\sigma^2 + a^2\sigma_0^2}.$$

Unrolling the GD recursion gives

$$\theta_{k+1}^{\mathrm{GD}} = r_{\mathrm{GD}}^{k+1}\theta_0 + \delta \frac{a}{\sigma^2 + a^2\sigma_0^2} y \frac{1 - r_{\mathrm{GD}}^{k+1}}{1 - r_{\mathrm{GD}}}.$$

There are a few remarks:

- Both EM and GD have the same fixed point $\theta^\star = \frac{y}{a}$ as $k \to \infty$, provided $|r_{\mathrm{GD}}| < 1$.
- EM has a fixed contraction factor r_{EM} which is independent of the iteration, while GD has a contraction factor r_{GD} which depends on the step size δ. By choosing δ appropriately so that $|r_{\mathrm{GD}}| < r_{\mathrm{EM}}$, GD can converge faster than EM in this case.

Example 5.5. Derive a gradient based MMLE method assuming that gradients are estimated with SNIS at every iteration with a sequence of proposal distributions $q_k(x)$.

Solution. The gradient based MMLE iterations are given by

$$\theta_{k+1} = \theta_k + \delta \nabla_\theta \log p_{\theta_k}(y).$$

Using Fisher's identity, we can write

$$\nabla_\theta \log p_{\theta_k}(y) = \mathbb{E}_{p_{\theta_k}(x|y)}[\nabla_\theta \log p_{\theta_k}(x, y)].$$

We can estimate this expectation with SNIS using the proposal distribution $q_k(x)$ as

$$\begin{aligned}
\nabla_\theta \log p_{\theta_k}(y) &= \int \nabla_\theta \log p_{\theta_k}(x, y) p_{\theta_k}(x|y) dx, \\
&= \int \nabla_\theta \log p_{\theta_k}(x, y) \frac{p_{\theta_k}(x|y)}{q_k(x)} q_k(x) dx, \\
&= \frac{\int \nabla_\theta \log p_{\theta_k}(x, y) \frac{p_{\theta_k}(x,y)}{q_k(x)} q_k(x) dx}{\int \frac{p_{\theta_k}(x,y)}{q_k(x)} q_k(x) dx}, \\
&\approx \frac{\frac{1}{N}\sum_{i=1}^N \nabla_\theta \log p_{\theta_k}(X_k^{(i)}, y) \frac{p_{\theta_k}(X_k^{(i)},y)}{q_k(X_k^{(i)})}}{\frac{1}{N}\sum_{i=1}^N \frac{p_{\theta_k}(X_k^{(i)},y)}{q_k(X_k^{(i)})}}, \\
&= \frac{\sum_{i=1}^N \mathsf{W}_{i,k} \nabla_\theta \log p_{\theta_k}(X_k^{(i)}, y)}{\sum_{i=1}^N \mathsf{W}_{i,k}}, \\
&= \sum_{i=1}^N \bar{\mathsf{w}}_{i,k} \nabla_\theta \log p_{\theta_k}(X_k^{(i)}, y),
\end{aligned}$$

where $X_k^{(i)} \sim q_k(x)$ i.i.d., $\mathsf{W}_{i,k} = \frac{p_{\theta_k}(X_k^{(i)},y)}{q_k(X_k^{(i)})}$, and

$$\bar{\mathsf{w}}_{i,k} = \frac{\mathsf{W}_{i,k}}{\sum_{j=1}^N \mathsf{W}_{j,k}}$$

are the normalised importance weights. Therefore, the gradient based MMLE method with SNIS gradient estimation is given by

$$\theta_{k+1} = \theta_k + \delta \sum_{i=1}^N \bar{\mathsf{w}}_{i,k} \nabla_\theta \log p_{\theta_k}(X_k^{(i)}, y).$$

Example 5.6. Consider the target density on $\theta \in [0,1]$ and $x = (x_1, \ldots, x_d) \in [0,1]^d$:

$$\pi(\theta, x) \propto \mathbf{1}_{[0,1]}(\theta) \prod_{i=1}^d \mathbf{1}_{[0,1]}(x_i).$$

Hence θ and x_i are independent uniforms under π, and the marginal target of interest is

$$\pi_\theta(\theta) \propto \mathbf{1}_{[0,1]}(\theta).$$

With a jump $\pm h$ proposal, compare the average acceptance probabilities of a joint sampler that updates θ and x together with a pseudo-marginal parameter-only sampler that updates only θ with an unbiased estimator of $\pi_\theta(\theta)$ with multiplicative noise as in the previous example.

Solution. We start with the joint sampler. Fix $h \in (0, \frac{1}{2}]$.

Joint sampler. At state (θ, x), propose

$$\theta' = \theta + \eta_0 h, \qquad x_i' = x_i + \eta_i h, \quad i = 1, \dots, d,$$

where $\eta_0, \eta_1, \dots, \eta_d$ are i.i.d. on $\{-1, +1\}$ with probability $1/2$ each. The proposal is symmetric, and π is constant on $[0, 1]^{d+1}$ and zero outside. Therefore

$$\alpha_{\text{joint}}((\theta, x), (\theta', x')) = \min\left(1, \frac{\pi(\theta', x')}{\pi(\theta, x)}\right) = \mathbf{1}\{\theta' \in [0, 1],\ x_i' \in [0, 1]\ \forall i\}.$$

So acceptance is exactly the probability that all proposed coordinates stay in the box.

At stationarity, $\theta, x_1, \dots, x_d$ are i.i.d. $\mathrm{Unif}(0, 1)$. For one generic coordinate $Z \sim \mathrm{Unif}(0, 1)$ with proposal $Z' = Z + \eta h$:

$$\mathbb{P}(Z' \in [0, 1] \mid Z = z) = \begin{cases} \frac{1}{2}, & z \in [0, h), \\ 1, & z \in [h, 1 - h], \\ \frac{1}{2}, & z \in (1 - h, 1]. \end{cases}$$

Hence

$$\begin{aligned} \mathbb{P}(Z' \in [0, 1]) &= \int_0^h \frac{1}{2}\, dz + \int_h^{1-h} 1\, dz + \int_{1-h}^1 \frac{1}{2}\, dz \\ &= \frac{h}{2} + (1 - 2h) + \frac{h}{2} = 1 - h. \end{aligned}$$

By independence across coordinates,

$$\bar{\alpha}_{\text{joint}} := \mathbb{E}_\pi\Big[\alpha_{\text{joint}}\Big] = (1 - h)^{d+1}.$$

Pseudo-marginal parameter-only sampler. Now update only θ with $\theta' = \theta + \eta h$, $\eta \in \{-1, +1\}$ equiprobable, and use an unbiased multiplicative estimator

$$\widehat{\pi_\theta}(\theta) = \pi_\theta(\theta)\, W, \qquad W = \begin{cases} 1 + c, & \text{with probability } 1/2, \\ 1 - c, & \text{with probability } 1/2, \end{cases} \qquad 0 < c < 1.$$

Since $\mathbb{E}[W] = 1$, this is unbiased. At stationarity,

$$\mathbb{P}(W = 1 + c) = \frac{1 + c}{2}, \qquad \mathbb{P}(W = 1 - c) = \frac{1 - c}{2},$$

while

$$\mathbb{P}(W' = 1 + c) = \mathbb{P}(W' = 1 - c) = \frac{1}{2}.$$

Hence

$$\alpha_{\mathrm{PM}}(\theta,\theta') = \min\left(1, \frac{\widehat{\pi_\theta}(\theta')}{\widehat{\pi_\theta}(\theta)}\right) = \mathbf{1}_{\{\theta'\in[0,1]\}} \min\left(1, \frac{W'}{W}\right),$$

because $\pi_\theta(\theta) = \mathbf{1}_{[0,1]}(\theta)$. At stationarity, $\theta \sim \mathrm{Unif}(0,1)$, so

$$\mathbb{P}(\theta' \in [0,1]) = 1 - h.$$

Also,

$$\xi(c) := \mathbb{E}\left[\min\left(1, \frac{W'}{W}\right)\right] = \frac{1+c}{4}\cdot 1 + \frac{1+c}{4}\cdot\frac{1-c}{1+c} + \frac{1-c}{4}\cdot 1 + \frac{1-c}{4}\cdot 1 = 1 - \frac{c}{2}.$$

Therefore

$$\bar{\alpha}_{\mathrm{PM}} := \mathbb{E}[\alpha_{\mathrm{PM}}] = (1-h)\,\xi(c) = (1-h)\left(1 - \frac{c}{2}\right).$$

Comparison.

$$\bar{\alpha}_{\mathrm{joint}} = (1-h)^{d+1}, \qquad \bar{\alpha}_{\mathrm{PM}} = (1-h)\left(1 - \frac{c}{2}\right).$$

Hence

$$\frac{\bar{\alpha}_{\mathrm{PM}}}{\bar{\alpha}_{\mathrm{joint}}} = \left(1 - \frac{c}{2}\right)(1-h)^{-d}.$$

So for fixed $h \in (0, \frac{1}{2}]$ and fixed noise level $c \in (0,1)$, joint-sampling acceptance decays exponentially in d, while pseudo-marginal acceptance is dimension-free in d (it only pays the noise factor $1 - \frac{c}{2}$ and the one-dimensional boundary factor $1 - h$).